

CMOS VLSI Design Project

Design and Analysis of a 2-Input CMOS AND Gate

Using Cadence Virtuoso

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1 Introduction

The objective of this project is to design, implement, and verify a two-input CMOS AND gate using the Cadence Virtuoso design environment. The project includes schematic design, symbol generation, physical layout design, design-rule checking (DRC), layout-versus-schematic (LVS) verification, parasitic extraction using RCX, and post-layout performance analysis.

A two-input AND gate produces a logic HIGH output only when both input signals are HIGH. In this project, the AND operation is implemented using a two-input NAND gate followed by a CMOS inverter.

The overall logic expression is

$$Y = A \cdot B \quad (1)$$

where A and B are the two input signals and Y is the output.

2 Objectives

The major objectives of this project are:

- To design a two-input CMOS AND gate in Cadence Virtuoso.
- To create the transistor-level schematic of the logic circuit.
- To generate a reusable symbol for the AND gate.
- To design the physical layout of the circuit.
- To verify that the layout satisfies all fabrication design rules using DRC.
- To confirm that the layout and schematic are electrically equivalent using LVS.
- To perform RC parasitic extraction using RCX.
- To perform post-layout simulation using the extracted circuit.
- To calculate the average power consumption.
- To determine the propagation delay of the AND gate.

3 Theory of AND Gate

A two-input AND gate follows the Boolean expression

$$Y = A \cdot B \quad (2)$$

The truth table of the AND gate is given in Table 1.

Table 1: Truth Table of a 2-Input AND Gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

A CMOS AND gate can conveniently be implemented using a NAND gate followed by an inverter.

For the NAND stage,

$$X = \overline{A \cdot B} \quad (3)$$

The output of the NAND gate is then applied to an inverter:

$$Y = \overline{X} \quad (4)$$

Therefore,

$$Y = \overline{\overline{A \cdot B}} = A \cdot B \quad (5)$$

Hence, the overall circuit performs the required AND operation.

4 Circuit Design

4.1 Schematic Design

The two-input AND gate was designed in Cadence Virtuoso using a two-input NAND gate followed by an inverter. The two input terminals are labeled A and B , while the final output terminal is labeled Y .

The NAND gate initially generates the complemented AND result. The inverter connected to its output restores the required AND logic.

The schematic used in the project is shown in Figure 1.

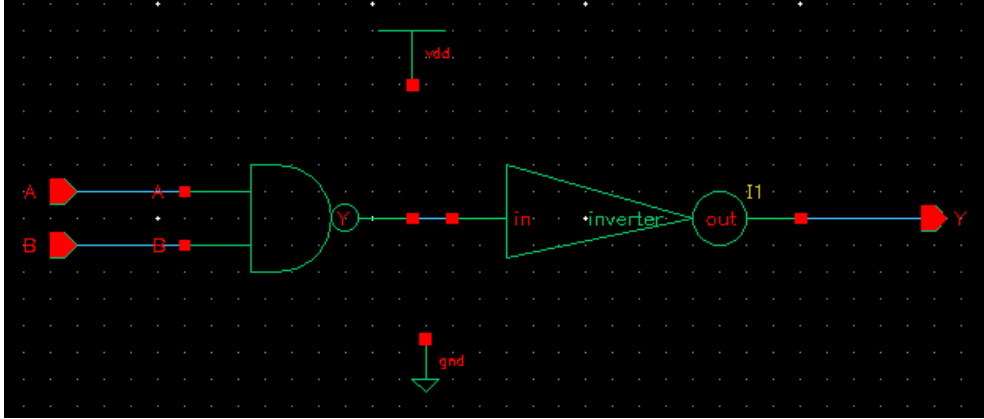


Figure 1: Schematic of the two-input AND gate in Cadence Virtuoso.

The internal operation can be represented as

$$A, B \xrightarrow{\text{NAND}} \overline{AB} \xrightarrow{\text{Inverter}} AB. \quad (6)$$

4.2 AND Gate Symbol

After successful schematic creation and verification, a symbol was generated for the AND gate. The symbol provides two input pins, A and B , and one output pin, Y .

The generated symbol is shown in Figure 2.

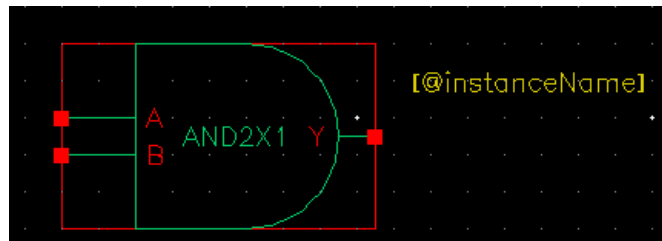


Figure 2: Generated symbol of the two-input AND gate.

Creating a symbol allows the designed AND gate to be used conveniently as a reusable component in larger hierarchical circuits.

5 Physical Layout Design

The physical layout of the AND gate was developed based on the transistor-level schematic. PMOS and NMOS transistors were placed and interconnected using the appropriate diffusion, polysilicon, metal, contact, and well layers.

The layout was designed carefully so that:

- the physical connectivity matches the schematic,

- the PMOS devices are connected toward the power supply,
- the NMOS devices are connected toward ground,
- the input signals A and B are routed to the required transistor gates,
- the NAND output is connected to the inverter input,
- the final output is routed to terminal Y , and
- fabrication design rules are maintained.

The completed layout is shown in Figure 3.

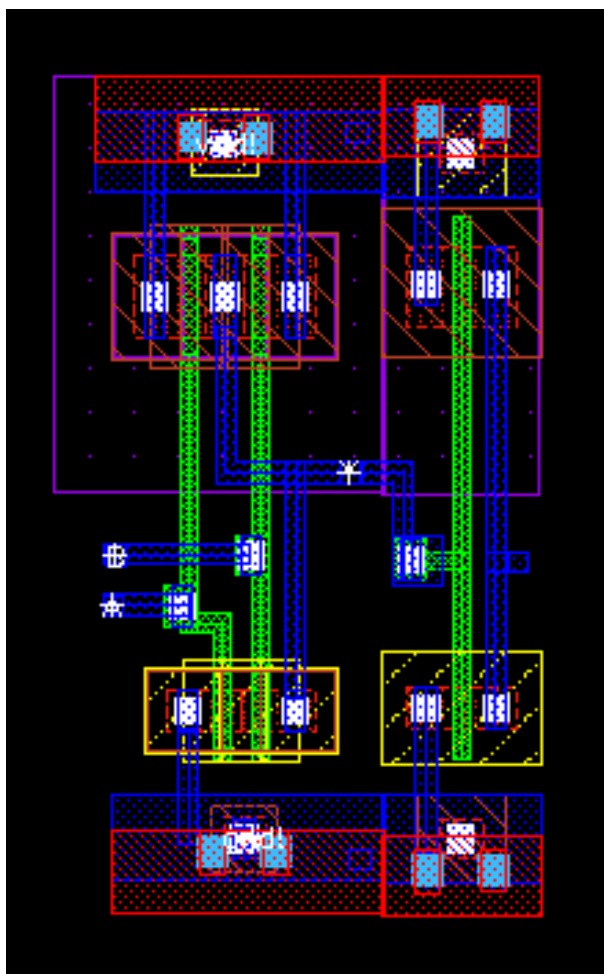


Figure 3: Physical layout of the CMOS AND gate.

6 Design Rule Check (DRC)

Design Rule Check, or DRC, is used to determine whether the physical layout satisfies the fabrication rules associated with the selected technology.

Typical rules checked by DRC include:

- minimum metal width,
- minimum metal spacing,
- polysilicon spacing,
- diffusion spacing,
- enclosure requirements,
- contact and via rules, and
- well and active-area requirements.

The AND gate layout was checked using the DRC tool in Cadence.

DRC Result: PASSED

The completed AND gate layout satisfied the required design rules.

Therefore, the physical layout is considered valid with respect to the technology design rules used in the project.

7 Layout Versus Schematic (LVS)

Layout Versus Schematic verification was performed after successful DRC verification.

LVS compares the devices and electrical connections extracted from the physical layout with those present in the original schematic.

The verification checks parameters such as:

- number of MOS transistors,
- transistor type,
- source and drain connectivity,
- gate connectivity,
- input and output pins,
- power connections, and
- overall circuit topology.

For this project, the LVS comparison was completed successfully.

LVS Result: PASSED

The physical layout matches the original schematic electrically.

This confirms that no unwanted electrical connectivity error was introduced during layout design.

After successful DRC and LVS verification, RC extraction was performed. RCX extracts parasitic resistance and capacitance associated with the physical interconnections and device geometries.

The extracted layout of the AND gate is shown in Figure 4.

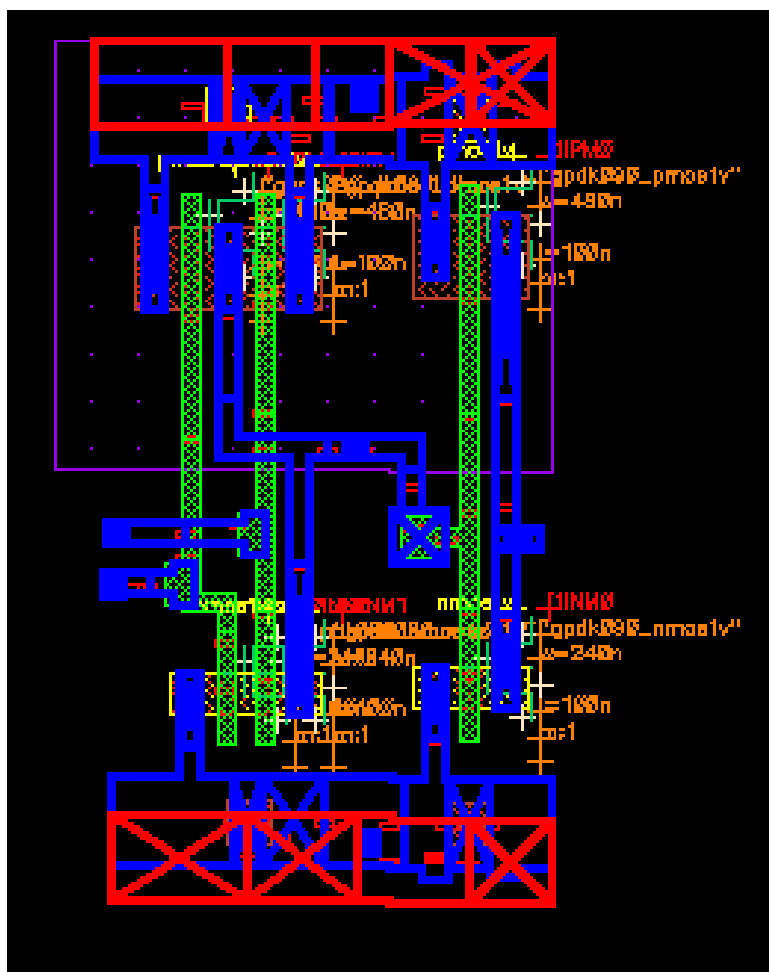


Figure 4: RCX/extracted view of the AND gate layout.

RCX: Successfully Completed

The extracted circuit was used for post-layout simulation and performance calculation.

After parasitic extraction, post-layout simulation was performed using the extracted circuit. Post-layout simulation provides a more realistic estimate of circuit performance

because the parasitic resistance and capacitance of the physical layout are included.

The main performance parameters evaluated in this work were:

1. Average power consumption
2. Propagation delay

10 Average Power Calculation

The average power consumption was determined from the transient simulation results using the Cadence Virtuoso Calculator.

The calculator expression used to determine average power was based on the average of the simulated power waveform:

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (7)$$

where $p(t)$ represents the instantaneous power and T represents the measurement interval.

The Cadence calculator result is shown in Figure 5.

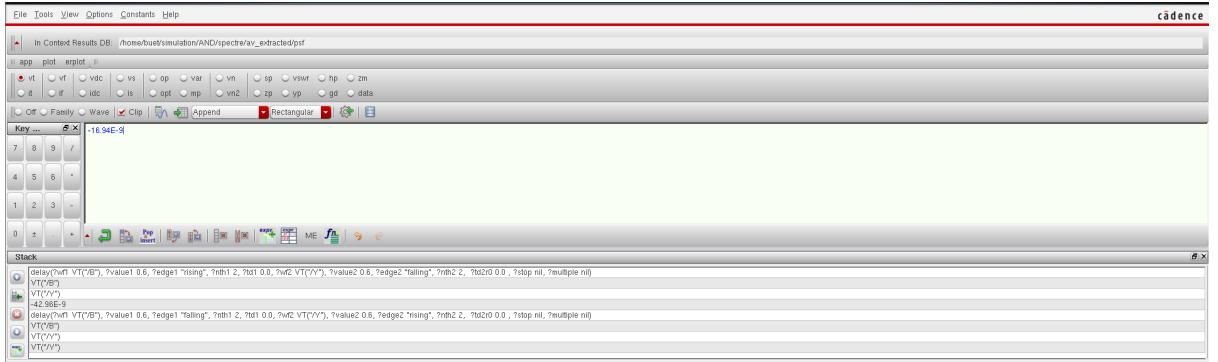


Figure 5: Average power calculation using the Cadence Calculator.

From the simulation, the obtained average power is approximately

$$P_{\text{avg}} = 5.862 \times 10^{-6} \text{ W}. \quad (8)$$

Therefore,

$$P_{\text{avg}} \approx 5.862 \mu\text{W} \quad (9)$$

This result represents the average power consumed by the simulated AND gate for the applied input conditions and simulation setup.

11 Propagation Delay Calculation

Propagation delay represents the time required for a change at the input of a digital gate to produce the corresponding change at the output.

For a CMOS gate, two propagation-delay components are normally considered:

$$t_{pHL} = t_{\text{output falling}} - t_{\text{input triggering}} \quad (10)$$

and

$$t_{pLH} = t_{\text{output rising}} - t_{\text{input triggering}}. \quad (11)$$

The average propagation delay is generally expressed as

$$t_p = \frac{t_{pHL} + t_{pLH}}{2}. \quad (12)$$

A switching threshold of approximately 0.6 V was used in the Cadence Calculator expressions to locate the relevant input and output transitions.

The delay calculation is shown in Figure 6.



Figure 6: Propagation-delay measurement using the Cadence Calculator.

The screenshot contains delay calculations between the selected transition of input B and the corresponding transition of output Y .

Note: The calculator screenshot displays signed time differences. Before reporting a final propagation-delay value, the input trigger edge and corresponding output target edge should be selected so that the calculated propagation delay represents the physical positive time interval between the cause and the resulting output transition.

12 Verification and Performance Summary

The major results obtained from this project are summarized in Table 2.

Table 2: Summary of AND Gate Design Results

Parameter	Result
Logic Function	$Y = A \cdot B$
Implementation	NAND2 followed by inverter
Schematic Design	Completed
Symbol Creation	Completed
Physical Layout	Completed
DRC	Passed
LVS	Passed
RCX / Parasitic Extraction	Completed successfully
Post-layout Simulation	Completed
Average Power	5.862 μ W
Propagation Delay	Calculated using Cadence Calculator

13 Discussion

The project successfully demonstrates the complete custom digital IC design flow for a two-input CMOS AND gate using Cadence Virtuoso.

Initially, the required Boolean operation was implemented using a NAND gate followed by an inverter. After verification of the schematic, a symbol was created so that the AND gate could be used as an independent hierarchical cell.

The physical layout was then developed according to the selected technology rules. Successful DRC verification indicates that the physical geometry of the design satisfies the required fabrication constraints.

The successful LVS result confirms that the electrical connectivity of the physical layout is equivalent to the designed schematic. This is important because a layout may satisfy fabrication rules while still containing incorrect electrical connections.

RC extraction was subsequently performed to include the effects of interconnect resistance and parasitic capacitance. The extracted circuit was then used for post-layout analysis.

The measured average power consumption was approximately

$$5.862 \mu\text{W}.$$

The propagation delay was also evaluated using transition-based measurements in the Cadence Calculator. Together, power and propagation delay provide useful information about the efficiency and speed of the implemented CMOS AND gate.

14 Conclusion

A complete two-input CMOS AND gate was successfully designed and verified using Cadence Virtuoso. The AND function was realized using a NAND gate followed by a CMOS inverter.

The project covered the major stages of the custom IC design flow, including schematic capture, symbol generation, physical layout, DRC, LVS, RC extraction, and post-layout simulation.

The layout successfully passed both DRC and LVS verification, confirming that the design satisfies the required physical design rules and matches the intended schematic connectivity. RCX was also successfully completed, enabling more realistic post-layout performance analysis.

The measured average power consumption of the circuit was approximately

$$\boxed{5.862 \mu\text{W}}.$$

Propagation-delay analysis was also performed using the Cadence Calculator. Overall, the project demonstrates the practical procedure for designing, verifying, and evaluating a CMOS digital logic gate from schematic level through physical implementation.

15 Software Used

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- DRC verification tool
- LVS verification tool
- RCX / parasitic extraction tool
- Cadence Virtuoso Visualization and Analysis Calculator
- Spectre simulator